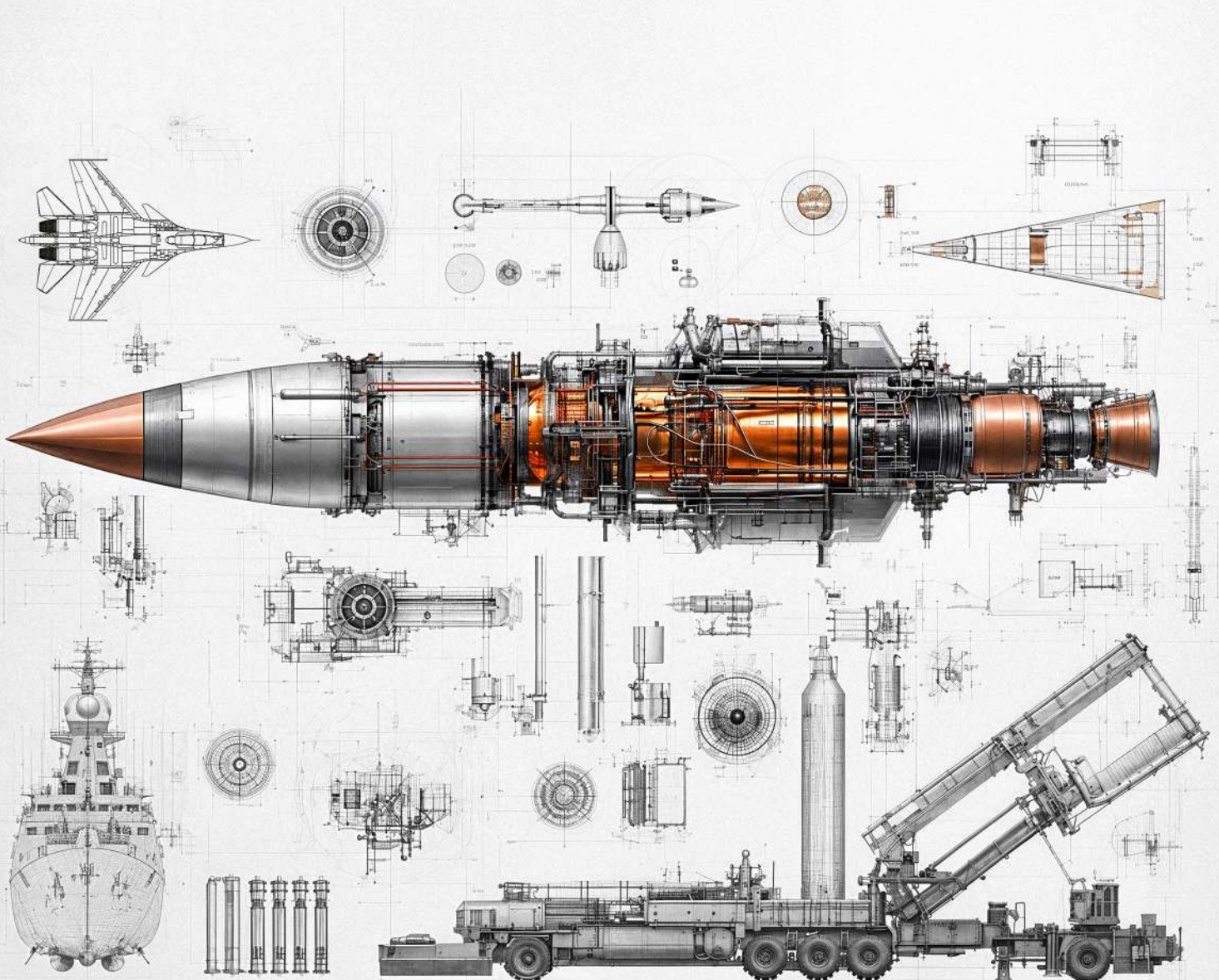


BUILDING INDIA'S *NEXT-GEN* AEROSPACE MANUFACTURING BASE

Advancing Indigenisation, Precision Engineering
and ***Strategic Manufacturing Partnerships***





BUILDING INDIA'S NEXT-GENERATION AEROSPACE MANUFACTURING BASE

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Strategic Manufacturing Partnerships

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3, San Martin Marg | Chanakyapuri | New Delhi - 110021
Tel: 011-24121764 | Fax: 011-66173415
E-mail: info@vifindia.org
Website: www.vifindia.org
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Foreword

As 2026 marks twenty-five years since the BrahMos missile's first successful flight, it is an opportune moment to reflect on what the BrahMos programme represents in India's quest for military self-reliance. Its success embodies India's growing confidence in its ability to conceive, develop, and deploy advanced military systems, at par with the best in the world, if not better. Over the past quarter-century, BrahMos has emerged as one of the country's most successful defence programmes, demonstrating how long-term vision, strong partnerships, technological innovation, and sustained national commitment can collectively strengthen India's security.

The BrahMos project was conceived with a clear objective: to provide the Indian Armed Forces with a highly capable and precision strike weapon system. Over the last twenty-five years, this objective has been realised with remarkable success. Today, BrahMos is among the most widely recognised missile systems in the world and has become an integral component of India's defence preparedness.

The significance of BrahMos extends beyond its military capabilities. It demonstrates India's ability to develop advanced technologies, build domestic industrial capacity, and reduce dependence on external sources for critical defence requirements. At a time when Aatmanirbharta has become a national imperative, the BrahMos programme offers valuable lessons for ongoing and future defence initiatives.

This success was not achieved overnight. It was made possible by sustained political support at the highest level, institutional commitment, dynamic leadership, and a clear understanding of the country's long-term security requirements. Engineers, scientists, military professionals, policymakers, and industry partners each played a vital role in turning an ambitious vision into a proven operational capability. Their collective efforts demonstrate what India can accomplish when diverse stakeholders work together towards a shared national objective.

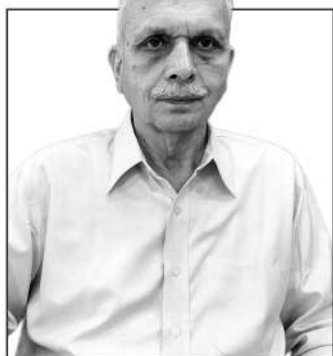
A particularly noteworthy achievement of the programme has been its successful operationalisation across all three services of the Indian Armed Forces. The induction of BrahMos into the Indian Army, Indian Navy, and Indian Air Force reflects both the missile system's versatility and the Services' confidence in it. Its integration across multiple platforms has strengthened India's military power and enhanced the Armed Forces' ability to respond effectively to evolving security challenges.



The programme has also played a significant role in strengthening India's defence industrial ecosystem. The steady increase in indigenous content, the growth of manufacturing expertise, and the development of a robust supply chain network have contributed to the creation of enduring industrial and technological capabilities within the country. These achievements have helped establish a stronger foundation for the indigenous production of advanced defence systems.

In recent years, BrahMos has also notched up significant export orders, adding a new dimension to its success. Its entry into international markets reflects the growing confidence of partner countries in Indian defence technologies. It underscores India's transformation from a major defence importer into a producer of advanced military capabilities and a credible participant in the global defence market.

This Policy Brief, ***Building India's Next-Generation Aerospace Manufacturing Base***, arrives at an important juncture. As India seeks to strengthen further its defence and technological capabilities, the lessons from the BrahMos experience remain highly relevant. The programme demonstrates the value of strategic vision, sustained investment in indigenous capabilities, close collaboration among stakeholders, and long-term commitment to national objectives. The story of BrahMos is not merely a successful missile programme; it is a testament to what India can achieve through collective national effort.



Dr Arvind Gupta

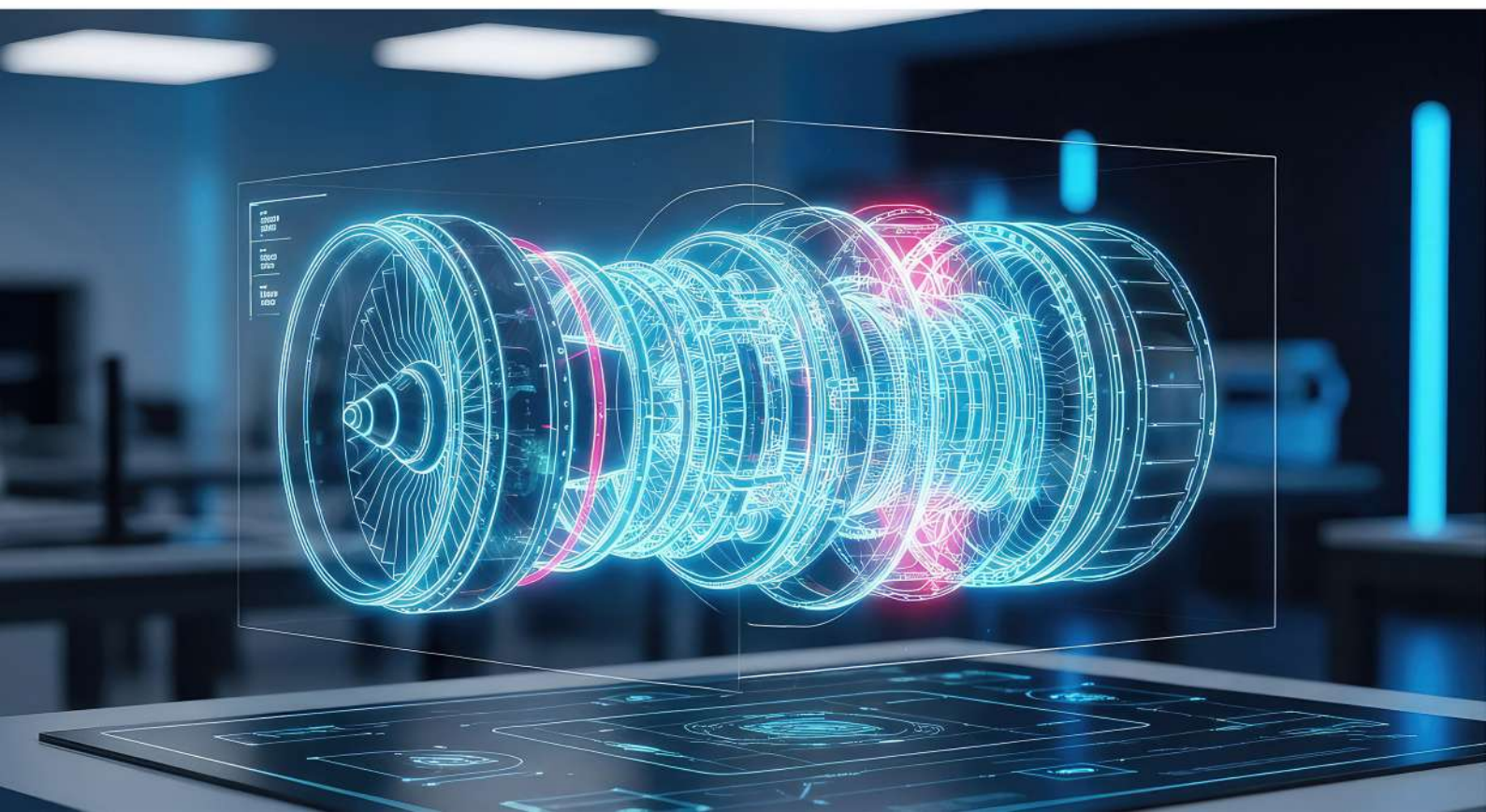
Director, Vivekananda International Foundation
New Delhi, June 2026

Executive *Summary*

India's aerospace and defence manufacturing ecosystem is entering a decisive phase of growth. As India seeks to build sovereign capabilities in aero-engines, propulsion systems, radar, electronic warfare systems, precision-strike systems and next-generation aerospace platforms, lessons from the 25-year evolution of the BrahMos programme offer an important blueprint for capability creation. The ongoing review of the Defence Acquisition Procedure (DAP 2026)¹ presents a timely opportunity to institutionalise procurement, industrial and technology development reforms that can accelerate India's transition from licensed production to indigenous design, development and manufacturing leadership.

Legacy businesses face structural challenges in participating effectively in aero engine and critical systems programmes because the procurement ecosystem disproportionately prioritises lowest-cost bidding over demonstrated technical capability and proven execution. Existing mechanisms, such as MSME preferences and reverse auctions, further skew competition, enabling underqualified players to secure complex projects through aggressive underbidding, often resulting in execution risks, delays and long-term cost inefficiencies. Limited weightage for technical evaluation, coupled with policy rigidity and audit-driven decision constraints, restricts procurement authorities' ability to prioritise capability-led selection, undermining the growth of credible, experienced industry players and slowing the development of a robust domestic aerospace manufacturing ecosystem.

The BrahMos experience offers important lessons not only for missile development but also for India's broader aerospace manufacturing ambitions, spanning critical propulsion technologies, advanced materials, mission-critical electronics, and other military capabilities. As India reviews DAP 2026 and advances its self-reliance agenda, these lessons provide a practical framework for building globally competitive and strategically resilient aerospace capabilities.



Key Findings

01

The BrahMos programme shows that building a strategic national capability requires more than technology acquisition. Sustained political commitment, mission-oriented governance, structured technology absorption, long-term industrial partnerships, and assured procurement demand were equally critical to its success.

02

India's aerospace ambitions require a shift from licensed production to sovereign capability development. Building indigenous competencies in propulsion systems, advanced materials, electronics, guidance systems and precision manufacturing will be essential to reducing long-term import dependence.

03

The development of a robust aerospace manufacturing ecosystem depends on deep integration among government, DRDO, industry, MSMEs and academia. The BrahMos experience illustrates how collaborative development models can accelerate indigenisation and strengthen domestic industrial capacity.

04

India's Defence Acquisition Procedure must evolve to recognise and reward demonstrated capability. DAP 2026 offers an opportunity to institutionalise capability based acquisition models that prioritise technology maturity, quality, performance and lifecycle value, alongside cost considerations.

05

India's defence production has grown significantly in recent years. However, India remains among the world's largest arms importers, underscoring the need to deepen indigenous capabilities in frontier technologies and advanced weapon systems.



Drawing on the BrahMos experience, this paper outlines a national playbook for developing aerospace capabilities, centred on co-development, indigenous innovation, industrial ecosystem development, sovereign supply chains and differentiated acquisition frameworks for strategic technologies.

The objective is not merely to produce advanced platforms but to build enduring national capabilities that strengthen India's technological sovereignty and global competitiveness.

The Playbook

This policy brief proposes a targeted policy advocacy agenda for DAP 2026, focused on three interconnected reforms:

01 | Capability-Weighted Evaluation

Mandating a minimum 70 per cent technical weighting in Request for Material Quotation (RMQ) processes for aero engine and critical aerospace system tenders, ensuring that demonstrated expertise and proven execution drive selection rather than price alone.

02 | Realistic Pricing Thresholds

Enforcing automatic disqualification of bids priced below 15 per cent of the established base price, thereby eliminating the structural incentive for underqualified players to win contracts through aggressive underbidding.

03 | Strengthened Institutional Oversight

Enabling the Ministry of Defence Production's engagement in procurement decision-making for complex aerospace programmes, providing a capability-informed check on audit-driven conservatism that currently constrains merit-based selection.



The DAP 2026 review is not merely an administrative exercise. It is an opportunity to align India's procurement architecture with its strategic ambitions building an aerospace manufacturing ecosystem that prioritises technical excellence, enforces realistic pricing, and ensures that the most capable and credible players can participate in and prosper from programmes that matter for national security.

Section 1

Context: India's Pursuit of Defence Self-Reliance

1.1 | India's Evolving Defence Industrial Trajectory

For the first six decades after independence, India's defence production base was dominated by Defence Public Sector Undertakings (DPSUs) and ordnance factories operating under licence from foreign technology partners. Volume production of standard platforms was possible, but frontier-technology design and development were not. The result was a structural dependency on foreign arms suppliers that successive governments acknowledged but struggled to address systematically.²

The policy landscape has transformed since 2014, with a sharp acceleration following the Galwan Valley clash in June 2020 and Russia's special military operations in Ukraine in February 2022, which exposed the risks of concentrated supply-chain dependence. The Aatmanirbhar Bharat (Self-Reliant India) initiative placed defence indigenisation at the centre of the national strategy. Aggregate production results have been striking. According to the Ministry of Defence's November 2025 report on Defence Aatmanirbharta, India's domestic defence production reached a record ₹1.54 lakh crore in FY 2025–26, a 174 per cent increase from ₹46,429 crore in FY 2014–15. Defence exports rose from less than ₹1,000 crore in 2014 to ₹38,424 crore (approximately US\$4.3 billion) in FY 2025–26, reaching over 80 countries³. The Ministry signed a record 193 contracts worth ₹2,09,050 crore in 2024–25, of which 177 contracts valued at ₹1,68,922 crore went to the domestic industry.⁴

Yet the structural challenge remains. India remained the world's second-largest importer of major arms in 2021–25, accounting for 8.3 per cent of global imports, even though its global share fell to 8.2 per cent from 9.3 per cent for the period 2016–2020, and its imports fell by 4.0 per cent between 2016–20 and 2021–25, partly due to India's growing ability to design and produce its own weapons⁵. Russia remained India's largest supplier, though its share fell from 70 per cent in 2011–15 to 48 per cent in 2021–25⁶. India was the largest recipient of arms from both France (24 per cent of French exports) and Israel (29 per cent of Israeli exports).⁷

This paradox, with record production numbers alongside continued import dependence, highlights the distinction between volume manufacturing and frontier-technology development. India can build nuclear submarines, armoured vehicles, and helicopters; yet it has struggled to develop high-performance aero-engines, advanced radar seekers, and certain precision manufacturing capabilities⁸. BrahMos, with its 85 per cent indigenous content target in 2026⁹, is a notable exception to this pattern.

Section 1

Context: India's Pursuit of Defence Self-Reliance

1.2 | Strategic Importance of Precision-Strike Capability

India's security environment is defined by a two-front challenge, making precision long-range strike capability a must-have. To India's north and east, China's People's Liberation Army (PLA) has invested heavily in precision strike and anti-access/area-denial (A2/AD) systems and has rapidly expanded its nuclear & conventional missile arsenals¹⁰. To the west, Pakistan possesses a nuclear deterrent and has historically supported cross-border terrorism. In this environment, the ability to hold adversary high-value targets, including air bases, command-and-control centres logistics nodes, and naval assets, at risk from survivable, mobile platforms is essential.¹¹

BrahMos uniquely addresses this requirement. Flying at Mach 2.8–3.0, nearly three times the speed of sound¹², it reduces adversary reaction time to seconds, defeating most existing air defence systems. Its fire-and-forget guidance, terrain-hugging flight profile, and low radar cross-section make interception extremely difficult. The missile's tri-service versatility ensures survivable launch platforms. With range progressively extended from 290 km (pre-MTCR) to 450 km and ultimately to 800 km, variants are now under development¹³, enabling BrahMos to hold adversary assets deep inside their territory at risk from positions well within Indian borders.

This combination of speed, range, precision, and platform versatility places BrahMos among the world's most capable operational cruise missile systems. The US Tomahawk, by comparison, flies at approximately Mach 0.8¹⁴, while the French Apache missile reaches around Mach 1.0¹⁵. China's YJ-18 is subsonic during much of its cruise phase before accelerating during terminal engagement. BrahMos is distinguished by its sustained supersonic performance, progressive range enhancements¹⁶, tri-service deployment across land, sea and air platforms, and a steadily increasing degree of indigenous content, a combination that few comparable missile systems have achieved simultaneously.

1.3 | Lessons from Capability Gaps and the Kargil Inflection

The Kargil War of 1999 was the strategic catalyst that gave BrahMos, whose inter-governmental agreement (IGA) was signed just months before the conflict, its institutional momentum. The war exposed critical deficiencies: the absence of rapid-response precision deep-strike capability, limited real-time ISR, and the constraints of conducting high-altitude operations without adequate indigenous logistics or ordnance. India fought and won, but the cost in lives and diplomatic capital underscored the price of capability gaps accumulated over decades of import dependence.

The evolution of India's aero-engine ecosystem reflects the complexity and long-term nature of propulsion development. The Kaveri engine programme, initiated to support the Tejas Light Combat Aircraft¹⁷, contributed significantly to India's indigenous knowledge base in gas turbine engineering, materials science, testing infrastructure, and propulsion research. These institutional and technological foundations are now informing a new generation of propulsion initiatives, including the proposed 120 kN-class engine for the Advanced Medium Combat Aircraft (AMCA) in collaboration with France's Safran under an Indian intellectual property framework¹⁸.

Section 1

Context: India's Pursuit of Defence Self-Reliance

The broader lesson from both BrahMos and India's propulsion initiatives is that sustained institutional commitment, structured co-development, and long-term industrial partnerships are essential to building strategic technological capability in frontier domains.

1.4 | The Policy Landscape: DAP 2020 and the Year of Reforms

India's Defence Acquisition Procedure 2020 (DAP 2020)¹⁹, which superseded the DPP 2016²⁰, introduced meaningful reforms: prioritisation of the 'Buy Indian-IDDM' (Indigenously Designed, Developed and Manufactured) category, the Strategic Partnership Model, expanded Make-I and Make-II provisions, leasing as a procurement category, and a Technology Development Fund. The FDI cap in defence was raised to 74 per cent under the automatic route. These were genuine improvements.

Nevertheless, DAP 2020 has been criticised for applying a uniform procurement logic across systems of vastly different complexity, prompting the Ministry of Defence to initiate a comprehensive review in 2025, which was declared the 'Year of Reforms'²¹. The review aims to align the procedure with evolving national policies, including Aatmanirbhar Bharat, to integrate emerging technologies, and to streamline procurement processes²². This ongoing reform provides the institutional opening for the recommendations in this policy brief, specifically, the creation of a differentiated procurement pathway for high-technology strategic programmes.



Section 2

BrahMos at 25: Evolution of a Strategic Programme

2.1 | Origins and Strategic Rationale

The genesis of BrahMos lies in a convergence of strategic necessity and technological opportunity in the final years of the Cold War. Russia's NPO Mashinostroyeniya (NPOM) had developed world-leading expertise in supersonic ramjet propulsion through the P-800 Oniks programme²³. Following the Soviet Union's dissolution, this capability required a partner to sustain and commercialise it. India, meanwhile, had built a nascent but significant capability base through the Integrated Guided Missile Development Programme (IGMDP) led by Dr APJ Abdul Kalam: inertial navigation systems, mission software, and mobile launcher architecture for the Prithvi, Agni, Trishul, Akash and Nag missile families²⁴.

As BrahMos Aerospace's official profile states: **"DRDO had developed crucial systems like inertial navigation systems, mission software, mobile launchers for Prithvi and Agni missiles. NPOM had expertise in the area of ramjet engines together with a number of technologies for space systems, launch vehicles and cruise missiles. Thus, the combination of the capabilities and competencies of both institutions gave birth to a world-class product, BRAHMOS, in a short time²⁵."**

The Inter-Governmental Agreement was signed on 12 February 1998, with authorised capital of US\$250 million (India: 50.5%, Russia: 49.5%)²⁶. The asymmetric shareholding was deliberate: India secured majority control while Russia retained a material stake that aligned its incentives with the programme's success. BrahMos Aerospace Private Limited was incorporated as a joint venture company, operating under commercial law rather than government procurement rules, a structural decision whose implications for programme governance proved decisive.

2.2 | Programme Evolution: From First Launch to Combat Validation

The first successful BrahMos launch occurred on 12 June 2001 from a land-based launcher at the Interim Test Range, Chandipur, Odisha, a milestone that established India as one of only a handful of nations capable of designing and fielding a supersonic cruise missile²⁷. Progressive development, testing, & induction across the three services followed in a disciplined sequence:

Section 2

BrahMos at 25: Evolution of a Strategic Programme

Year/Period	Milestones
1998	Inter-Governmental Agreement signed (12 Feb); BrahMos Aerospace JV established
2001	First successful launch at Chandipur, Odisha
2005	Ship-based variant inducted into the Indian Navy (INS Rajput)
2007	Land-based variant inducted into the Indian Army; First regiment of 67 Block-I missiles
2011	Second Army regiment; Block-II with an advanced seeker, commissioned
2016	India joins the Missile Technology Control Regime (MTCR); Range development begins
2017	First successful air-launch test from Su-30MKI
2020	Air-launched BrahMos-A inducted into the Indian Air Force
2022	Philippines deal signed: US\$375 million for three coastal defence batteries
2023	Indigenous seeker successfully tested
2024	First battery of BrahMos delivered to the Philippines
2025	Operation Sindoor: first combat use; targets struck in Pakistan Lucknow Integration & Testing Facility inaugurated (₹300 crore, 80 ha)
2025	First consignment dispatched from Lucknow facility (18 October)
2025	Indigenous solid rocket booster flight-tested successfully
2026 Target	85% indigenous content milestone; BrahMos-NG development advancing

Section 2

BrahMos at 25: Evolution of a Strategic Programme

2.3 | Tri-Service Integration and Platform Expansion

BrahMos's deployment across all three services, each requiring distinct integration engineering, represents a significant operational achievement. The Navy inducted the ship-launched version first in 2005, equipping INS Rajput and progressively extending to modern destroyers, frigates, and corvettes. In March 2024, the Navy concluded the largest-ever indigenous missiles contract with M/s BrahMos Aerospace Pvt Ltd. for the supply of enhanced-range BrahMos missiles²⁸. The Army operates the land-based mobile autonomous launcher variant across multiple regiments deployed along both the northern (China-facing) and western (Pakistan-facing) fronts²⁹. In 2025, the Defence Acquisition Council approved additional procurement of extended-range BrahMos variants³⁰.

The integration of BrahMos with the Su-30MKI deserves particular attention as a case study in domestic engineering capability. When India sought to integrate the 2.5-tonne BrahMos on the Su-30MKI airframe, Russia's Sukhoi corporation quoted approximately US\$200 million (approximately ₹13 billion) for the work. Hindustan Aeronautics Limited (HAL) undertook the same integration domestically at a cost of approximately ₹0.8 billion (approximately US\$9.6 million at prevailing rates), roughly one-fifteenth of the Russian estimate³¹. A growing number of Su-30MKI aircraft have been modified to carry BrahMos under ongoing modernisation efforts.

2.4 | Indigenisation: A 25-Year Journey

The indigenisation trajectory of BrahMos, from 13 per cent at inception to approximately 83 per cent by 2025³², is the most consequential dimension of the programme for India's broader strategic manufacturing agenda. This progression was not automatic; it required deliberate, sustained effort from BrahMos Aerospace, its Indian supply chain, and the government.

The indigenisation journey of BrahMos has accelerated significantly over the past decade, reflecting the steady expansion of India's defence-industrial ecosystem and growing domestic capability in advanced subsystems. Throughout this journey, DRDO has remained the principal technology partner for the programme, supporting subsystem development, propulsion research, seeker technologies, testing, qualification, and future capability upgrades. The organisation's continuous engagement has enabled effective technology absorption across the industrial ecosystem, facilitating the progressive indigenisation of critical subsystems and laying the foundation for successive upgrades and future variants.

Successive milestones, including indigenous seekers, propulsion components, specialised materials, and electronic systems, have deepened Indian participation across the programme's value chain. Recent successful testing of indigenous subsystems demonstrates the maturity of the co-development and capability-absorption model underpinning the programme. A critical milestone was achieved in November 2025, when a flight test proved that the indigenous solid rocket booster matched the performance of the Russian original³³. Similarly, DRDO's successful development and testing of indigenous liquid ramjet fuel in 2024 eliminated another key import dependency and further strengthened India's propulsion technology base³⁴.

Section 2

BrahMos at 25: Evolution of a Strategic Programme

BrahMos Indigenisation at a Glance



The remaining 15 per cent of non-indigenous content comprises primarily the liquid ramjet engine core and certain specialised electronic components still sourced from Russia. DRDO is developing an indigenous liquid-fuelled ramjet engine, which, once inducted, will push the programme towards 90–95 per cent indigenisation, potentially making BrahMos one of the most deeply indigenised advanced missile systems in the world.

Section 2

BrahMos at 25: Evolution of a Strategic Programme

2.5 | The Lucknow Facility and Industrial Ecosystem

The inauguration of the BrahMos Integration and Testing Facility in Lucknow on 11 May 2025, National Technology Day, the 27th anniversary of India's Pokhran nuclear tests, was a deliberate symbolic choice. Built on 80 hectares of land provided free of cost by the Uttar Pradesh government, the facility cost ₹300 crore and is the country's largest BrahMos integration and testing centre^{35|36}. It produced its first consignment of missiles on 18 October 2025, with Defence Minister Rajnath Singh and Chief Minister Yogi Adityanath presiding.

The facility targets production of 80–100 BrahMos missiles annually, scaling toward 150 units as operations mature. It integrates booster sub-assemblies, avionics, propellant, and ramjet engine components, and is equipped to handle the BrahMos-NG (Next Generation) variant. The facility has generated approximately 500 direct and 1,000 indirect jobs. Defence Minister Singh noted at the inauguration that BrahMos is **"not just a weapon but a message of India's military strength"**³⁷.

The broader supply chain built around BrahMos now encompasses over 200 Indian public and private defence enterprises and institutions³⁸. Key private sector partners include: Godrej & Boyce (Godrej Aerospace), which supplied the first airframe assembly for the air-launched BrahMos-A variant and manufactures airframes for both BrahMos and MRSAM³⁹; HAL, which performed the Su-30MKI integration at a fraction of the foreign estimate⁴⁰; Larsen & Toubro, which manufactures launchers and integrated systems⁴¹; and PTC Industries, a Lucknow-based company that has been supplying specialised titanium components to BrahMos since 2019 and secured a major order for titanium castings in August 2025⁴². This ecosystem represents exactly the kind of deep, tiered industrial base that the concept note's authors identify as the foundation of strategic self-reliance.



Section 3

Building Strategic Capability: The BrahMos Approach

Understanding why BrahMos succeeded requires examining not only what was built but also how it was governed, partnered, and sustained. This chapter analyses the five structural pillars that made the programme a success, each with direct implications for India's next generation of strategic capability development.

3.1 | Institutional Design and Mission-Mode Governance

The most consequential structural feature of BrahMos Aerospace is its legal form: a private limited company incorporated under Indian company law, rather than a DRDO programme, a DPSU, or a Ministry of Defence project. This distinction determines the governance framework in which the programme operates.

A DRDO programme operates within an institutional culture optimised for scientific research, with progress measured by technical milestones and publications. A DPSU operates within a protected market environment where commercial discipline is often secondary to policy objectives. BrahMos Aerospace, as a JV company, combined the strategic mandate of a government programme with the governance discipline of a commercial entity: a board of directors accountable for outcomes, management empowered to make operational decisions without ministerial approval, and contractual relationships with industry partners that are legally enforceable.

This governance model enabled mission-mode execution. Programme leadership, with Dr A Sivathanu Pillai as founding CEO, followed by Dr Sudhir Mishra, Dr Atul Rane, and Dr Jaiteerth Joshi, maintained continuity of vision over 25 years. The programme was not subject to the leadership changes and policy reversals that have affected other major Indian defence programmes. The IGA signed in 1998 was extended indefinitely in 2014⁴³, signalling sustained commitment at the highest governmental level in both countries.

3.2 | Structured Technology Transfer with Genuine Absorption

BrahMos's indigenisation journey highlights the critical distinction between technology transfer and technology absorption. In a transactional technology transfer arrangement, the recipient receives manufacturing drawings, tooling specifications, and production know-how for a defined system. The transfer is complete when the recipient can build the system. In a genuine co-development partnership, the recipient participates in design decisions, understands the engineering rationale behind technical choices, and builds the capability to modify, improve, and extend the system independently.

Section 3

Building Strategic Capability: The BrahMos Approach

BrahMos began as a technology transfer arrangement; the initial missile was largely based on NPOM's Oniks technology. Over time, the programme evolved into genuine co-development, as Indian engineers came to understand the system's architecture well enough to design and build indigenous alternatives to imported components. The seeker, the solid rocket booster, the liquid ramjet fuel, and now the ramjet engine itself has been progressively indigenised through this process of absorption. Each milestone reflects not merely a manufactured component but a growing body of Indian engineering knowledge.

3.3 | Long-Term Public–Private Industrial Partnership

BrahMos was explicitly designed to build India's defence industrial base, not merely to produce a missile. From the programme's earliest stages, BrahMos Aerospace sought to engage Indian private-sector firms as partners in subsystem development, component manufacturing, and ultimately co-development. This marked a deliberate departure from the DRDO model of in-house development followed by technology transfer to DPSUs.

Aerospace business of Godrej Enterprises Group exemplifies this trajectory. The company began by supplying components and has since evolved into one of BrahMos' principal industrial partners, manufacturing airframe assemblies and other metallic subsystems for the missile programme. Publicly available information indicates that its role has expanded over time to include a wider range of precision-engineered structures and mission-critical hardware, reflecting the gradual movement of Indian private firms from component suppliers to strategic manufacturing partners. The advanced manufacturing capabilities developed through programmes such as BrahMos have also enabled Godrej to participate in other national aerospace initiatives, including actuator development and Kaveri-derivative aero-engine manufacturing projects.

HAL's partnership with BrahMos is an equally instructive case. Beyond the Su-30MKI integration, completed at one-fifteenth of the Russian cost estimate, HAL and BrahMos have formalised a long-term partnership to develop critical electronic systems and enhance indigenous content across all BrahMos variants. This marks the evolution from a transactional supplier relationship to a capability co-development partnership, precisely the model this paper recommends for replication.

3.4 | Supply Chain Depth and MSME Integration

The 200-plus enterprise ecosystem built around BrahMos includes a significant MSME tier. This reflects a deliberate strategy of supply chain deepening that distributes both production capability and risk across a wide base of Indian manufacturers.

PTC Industries, a Lucknow-based precision manufacturer, exemplifies MSME integration in the programme. The company began supplying specialised titanium components and mill forms to BrahMos in 2019. By August 2025, it had secured a ₹100 crore order for titanium castings critical

Section 3

Building Strategic Capability: The BrahMos Approach

to the missile's airframe structure, while also serving global aerospace clients, including Safran, Dassault Aviation, BAE Systems, and Israel Aerospace Industries. This dual capability, delivering defence-critical components for a strategic programme alongside global commercial aerospace supply, represents the quality and competence benchmark that BrahMos participation has established for Indian MSMEs.

The Defence Minister's remarks at the Lucknow facility inauguration explicitly linked the MSME dimension to the broader **Aatmanirbhar** vision: "PTC Industries Limited in Lucknow is starting titanium and super-alloy material plants. Apart from that, the foundation for seven additional critical projects is also being laid⁴⁴." This supply chain depth, anchored by a programme with

3.5 | Assured Demand and Strategic Patience

Perhaps the most underappreciated enabler of BrahMos's success is the sustained, uninterrupted procurement demand from all three Indian armed services over 25 years. Strategic self-reliance in deep-technology manufacturing cannot be achieved on commercially viable timescales without government-backed certainty of demand. Private sector firms will not invest in advanced manufacturing capabilities, engineering talent pipelines, and quality certification processes if they cannot project sustained demand.

BrahMos has benefited from multiple large procurement orders across the Army, Navy, and Air Force, with each round providing not only revenue but also the demand signal that enabled supply chain investment. Recent procurement approvals continue to provide sustained demand visibility for the programme and its supply chain. The Philippines export contract added an important international dimension to programme demand. The programme is working to replicate the deal in Vietnam (advanced discussions), Indonesia (deal concluded 2026), and 15 other interested nations.



Section 4

Operationalisation Of Capability

4.1 | Deployment Across Services and Platforms

BrahMos is now fully integrated into the operational doctrine and order of battle of all three Indian armed services, a level of institutional embedding that distinguishes it from most other domestically developed systems. The Navy has equipped virtually all front-line warships, including destroyers, frigates, and corvettes. The ship-launched variant gives the Navy a potent anti-ship and land-attack capability that can be delivered from multiple dispersed platforms simultaneously, complicating adversary targeting. The Navy's March 2024 single largest order for BrahMos missiles signals the continued deepening of the programme's presence across the service.

The Army's mobile autonomous launcher regiments, deployed along both the Line of Actual Control (China) and the Line of Control (Pakistan), give India a credible land-based deep-strike deterrent from survivable, mobile platforms. The regiments' mobility, mounted on 12×12 heavy-duty trucks, means adversary targeting is a complex and time-sensitive challenge. Extended-range variants, now reaching 450 km and with 800 km versions under trial, dramatically expand the operational reach of each regiment.

The Air Force's BrahMos-A integration on the Su-30MKI fleet of 40 aircraft, modified as of July 2024, with 84 more planned under Super Sukhoi⁴⁵, provides a stand-off deep-strike capability that can penetrate advanced air-defence systems. The range of the air-launched variant has already been extended to 450 km. It will increase to over 800 km, enabling India to strike high-value targets hundreds of kilometres inside adversary territory from aircraft operating from Indian airspace.

4.2 | Operation Sindoor: The Combat Validation

The decisive proof of BrahMos's operational value came on the night of 10 May 2025, when Indian Air Force Su-30MKI aircraft conducted Operation **Sindoor**, India's military response to the 22 April 2025 Pahalgam terrorist attack that killed 26 civilians. BrahMos missiles struck Pakistani air bases, including Noor Khan, causing significant damage to radar installations, command centres, and ammunition depots⁴⁶.

The strategic significance of Operation Sindoor extends beyond its immediate military impact. It was the first confirmed combat use of an Indian-manufactured supersonic cruise missile, and the first proof that a system with 70–80 per cent indigenous content could perform to specification under real warfighting conditions. The successful penetration of Pakistan's air

Section 4

Operationalisation of Capability

defences, which include Chinese-supplied HQ-9 surface-to-air missile systems, demonstrated BrahMos's capacity to defeat modern integrated air defence networks⁴⁷.

The export implications were immediate and dramatic. Within weeks, 17 countries had expressed interest in acquiring BrahMos⁴⁸. Indonesia moved rapidly to conclude a deal in March this year. Vietnam deal for a UD \$ 629 million Brahmos missile package, including logistical support, was announced in May 2026⁵⁰. The combat-proven BrahMos is no longer merely a deterrence symbol; it is a demonstrated conventional capability that potential adversaries must now account for in their force planning.

4.3 | Export Trajectory and Strategic Diplomacy

India's first major strategic missile export to the Philippines set an important precedent for future defence exports. The first battery was delivered in April 2024, the second in April 2025. The Philippines Marine Corps' acquisition of BrahMos for coastal defence reflects the mission critical imperative: the hostile posture of the PLA Navy and the China Coast Guard requires credible anti-ship deterrence that Manila's existing inventory could not provide.

The Philippines precedent has set a template for BrahMos exports: government-to-government agreement structure; operator training and maintenance packages; on-schedule deliveries; and demonstrated operational credibility following real-world deployment by the manufacturer nation. This template, combined with Operation Sindoor's combat validation, has transformed the BrahMos export pipeline.

The strategic logic driving buyers is clear. Nations facing Chinese maritime pressure in the South China Sea, the Philippines, Vietnam, Indonesia, and Malaysia need a credible supersonic anti-ship deterrent. Nations seeking to diversify from Russian suppliers amid supply chain concerns post-Ukraine are looking at BrahMos as a premium alternative to the Oniks. Nations investing in naval modernisation across the Gulf, South America, and Africa see BrahMos as a force multiplier at a competitive cost.



Section 4

Operationalisation of Capability

India's defence export target of

Approximately US\$6 billion

₹50,000 Crore by 2029

which roughly doubles the FY 2024–25 level, is achievable primarily through BrahMos and its derivatives. The Lucknow facility will play an important role in supporting future domestic and export requirements.

4.4 | Feedback Loops: Users and Developers

One of BrahMos's less-discussed institutional strengths is the structured feedback loop among operational users (the three services), DRDO, industry partners and the developer-producer (BrahMos Aerospace). Throughout the programme's evolution, continuous engagement among these stakeholders ensured that operational requirements, technology development priorities and future capability enhancements remained closely aligned. This collaborative approach enabled the programme to adapt rapidly to evolving user requirements while maintaining long-term technology development objectives.

This feedback loop has driven key programme decisions: the development of the Block-II variant with an advanced seeker; the Block-III steep-dive capability for targeting mountain positions; the extension of range following MTCR entry; and the air-launched BrahMos-A configuration. The programme's next-generation BrahMos-NG, targeting approximately 1.5 tonnes and Mach 3.5 for compatibility with the Tejas fighter and submarine torpedo tubes, is being driven by requirements for a lighter, smaller weapon that can be carried in larger numbers⁵¹.

Ongoing collaboration among users, developers, DRDO laboratories and industry partners created an iterative development ecosystem that enabled rapid upgrades and operational adaptation. This model offers a useful template for future aerospace and defence programmes, where sustained stakeholder engagement can significantly accelerate capability maturation.



Section 5

The Structural Challenge: Rethinking India's Defence Capability Model

5.1 | The Limits of Existing Procurement Approaches

The success of the BrahMos programme was not solely a function of technology. Equally important were the institutional, industrial and procurement frameworks that enabled long-term capability development. The programme benefited from mission-oriented governance, sustained stakeholder commitment, technology absorption mechanisms, industrial partnerships and assured demand over an extended period. These characteristics stand in contrast to many conventional acquisition models, which remain focused on transactional procurement outcomes rather than strategic capability development.

The problem arises when the same framework is applied to fundamentally different types of requirements: complex, high-technology systems that require decade-long engineering development, deep industrial partnerships, and sustained investment in capability rather than transactional procurement of a defined product. For these systems, the DAP framework's emphasis on competitive tendering, L1 (lowest technically compliant bid) selection, and defined procurement timescales creates three systemic pathologies:

Capacity Dilution

When complex systems requiring deep engineering collaboration are selected based on price competition, the procurement process prioritises cost minimisation over capability maximisation. Technology transfer obligations, which should drive genuine absorption, are treated as compliance requirements rather than strategic outcomes.

Ecosystem Fragmentation

Long-term industrial partnerships, which are essential for supply chain deepening and genuine technology absorption, are structurally discouraged in a competitive tendering framework where each procurement case is rebid, and there is no institutional mechanism for relationship continuity.

Timeline Distortion

Real engineering development of advanced systems is iterative, non-linear, and resistant to fixed procurement timelines. When development programmes are structured as procurement cases with defined deliverables, fixed-price contracts, and regulatory oversight designed for transactional purchasing, the result is either schedule slippage (as the programme is bent to procurement constraints) or capability compromise (as designs are frozen prematurely to meet contractual timelines).

Section 5

The Structural Challenge: Rethinking India's Defence Capability Model

Aerospace manufacturing differs fundamentally from conventional procurement. Advanced propulsion systems, avionics, precision-guided weapons, sensors and aerospace materials require long development cycles, specialised infrastructure, highly skilled workforces and sustained investment in research and testing. Such capabilities cannot be created through isolated procurement transactions; they require ecosystem-level development supported by long-term policy and industrial commitment.

5.2 | Building Long-Term Capability in Aero-Engine Development

India's aero-engine development efforts are among the most technologically ambitious components of its strategic self-reliance agenda. Initiated in the mid-1980s and formally sanctioned in 1989, the Kaveri programme sought to develop the GTX-35VS engine for the Tejas Light Combat Aircraft⁵². Over the decades, the programme has contributed to building domestic expertise in propulsion engineering, testing infrastructure, materials science, and gas turbine research capabilities that continue to support India's evolving aerospace ecosystem.

The broader experience of programmes such as BrahMos and Kaveri underscores the importance of institutional design, sustained industrial participation, and structured co-development frameworks for advancing complex strategic technologies. BrahMos Aerospace's empowered joint-venture structure, sustained industry participation, and integrated production ecosystem demonstrate a successful model for accelerating capability development and industrial absorption in frontier domains.

India's current aero-engine initiatives increasingly reflect a collaborative approach. The proposed GTRE-Safran partnership to develop a 120 kN-class engine for the Advanced Medium Combat Aircraft (AMCA)⁵³, under an Indian intellectual property framework and with expanding private-sector participation, represents an important step in strengthening India's long-term propulsion capabilities.

5.3 | Misalignment Between Capability Development And Acquisition Frameworks

The challenge is not simply one of procurement efficiency but of strategic intent. Acquisition frameworks designed primarily to secure the lowest compliant bid are often ill-suited to programmes with the objective of long-term capability creation. For strategic aerospace technologies, the acquisition process must support technology maturation, industrial capacity building, supply-chain development and continuous innovation, alongside cost discipline.

DAP 2020's 'Buy Indian-IDDm' category and the Strategic Partnership Model⁵⁴ represent efforts to shift the framework toward capability development. But these mechanisms sit within the same broader procurement architecture, with the same competitive tendering logic, the same L1 selection pressures, and the same timeline constraints.

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The Structural Challenge: Rethinking India's Defence Capability Model

A VIF analysis notes that the DAP

.... should not only allow exceptions for single-vendor scenarios, especially for indigenous technologies, but also incorporate policies that prioritise innovation-driven monopolies over regulatory monopolies with a firm focus on innovation rather than market capture⁵⁵.

5.4 | Structural Constraints in Scaling High-Technology Programmes

Four structural constraints limit India's ability to scale the BrahMos model across multiple strategic programmes simultaneously.

First, the absence of long-term fiscal commitment: Indian defence programmes typically receive annual budget allocations that are subject to revision, creating uncertainty that private-sector partners cannot plan for. BrahMos benefited from an IGA that committed both governments to the programme; replicating this commitment for domestic-only programmes requires explicit multi-year budget frameworks.

Second, the DRDO–industry interface: DRDO's laboratory culture prioritises research over industrial production. Technology developed within DRDO laboratories often does not transition smoothly to industry because the institutional incentives, processes, and relationships for production are absent. BrahMos Aerospace resolved this by being a production-oriented JV from the outset, with DRDO as a shareholder rather than the programme manager. This production-oriented JV model, with DRDO as a minority technology shareholder, needs to be replicated.

Third, the private sector's risk appetite: Indian private-sector defence firms are still relatively young. They lack the capital reserves, government relationships, and, in some cases, the technical depth to absorb the risk of long-duration, uncertain technology programmes without government backstops. The iDEX initiative of the MoD⁵⁶, the Technology Development Fund of DRDO⁵⁷, and the Strategic Partnership Model of the DAP are attempts to address this, but they remain underscaled relative to the challenge.

Fourth, supply-chain dependence remains a significant strategic vulnerability. Many critical aerospace materials, specialised electronics, sensors, semiconductor components and propulsion subsystems are still sourced from overseas. Building sovereign aerospace supply chains will require targeted industrial policies, long-term demand visibility and coordinated investment across multiple tiers of the manufacturing ecosystem.

Section 5

The Structural Challenge: Rethinking India's Defence Capability Model

5.5 | Aerospace Manufacturing And Defence Acquisition: Strategic Imperatives

To strengthen national aerospace manufacturing capabilities and enhance defence preparedness, the above arguments are summarised below.

Capability-Based Acquisition

Move beyond L1 procurement and adopt capability-based evaluation frameworks.

Sovereign Aerospace Supply Chains

Develop domestic capabilities in critical materials, electronics, sensors, composites and semiconductors.

Government-Industry Co-Development

Institutionalise collaborative development models.

Tiered Industrial Ecosystem

Strengthen MSMEs, startups and supplier networks.

Government as Anchor Customer

Provide long-term procurement visibility and continuity of demand.

Section 6

From Programme to Ecosystem: Scaling Strategic Capability

6.1 | Expanding the Industrial Ecosystem

The BrahMos industrial ecosystem, 200-plus enterprises, a dedicated production facility in Lucknow, and supply chain relationships spanning precision manufacturing, avionics, propulsion, and advanced materials, was built over 25 years of sustained programme demand. Replicating this depth of ecosystem for other strategic programmes requires both time and a deliberate institutional strategy.

India's two Defence Industrial Corridors in Uttar Pradesh and Tamil Nadu provide the geographic and infrastructural framework. The UP DIC, spanning Lucknow, Kanpur, Aligarh, Agra, Jhansi, and Chitrakoot, had attracted ₹9,145 crore in investment by October 2025, with 289 MoUs signed and a potential investment pipeline of ₹66,423 crore, according to the MoD's November 2025 report. The BrahMos facility is the DIC's most advanced and operationally significant anchor investment. For the corridor model to succeed in generating genuine strategic capability rather than manufacturing activity, it needs anchor investors with sustained procurement demand and the level of technological intensity that BrahMos represents.

The model for scaling is clear: identify 3–5 strategic capability priorities (propulsion, avionics, radar, electronic warfare and mission systems, hypersonics, advanced materials, underwater systems); establish a dedicated, production-oriented JV or equivalent entity for each priority; anchor each entity in a DIC node with support from the surrounding MSME ecosystem; and back each with multi-year procurement commitments from the armed services.

6.2 | Strengthening Co-Development and IP Frameworks

BrahMos's indigenisation journey has been constrained by the IP architecture of the original 1998 JV. Although India holds the majority stake in BrahMos Aerospace, the technical IP for certain core components, particularly the liquid ramjet engine, remains with NPOM, limiting India's ability to substitute indigenous alternatives without Russian concurrence. This is an inherent limitation of a JV model in which one partner contributes existing IP.

Future co-development arrangements must be structured with more explicit IP provisions from the outset. India's approach to the AMCA engine co-development with France's Safran is under full Indian IP ownership. This explicit IP-first approach, if properly structured and enforced, represents an evolution beyond the BrahMos model and should be institutionalised as the standard for all future co-development arrangements.

Section 6

From Programme to Ecosystem: Scaling Strategic Capability

6.3 | Advancing Indigenisation in Critical Technologies

Three critical technology domains present the most urgent indigenisation requirements: propulsion systems, advanced seekers and guidance, and hypersonic propulsion.

On propulsion, the BrahMos programme's development of an indigenous liquid ramjet fuel, tested by DRDO in 2024, and the successful flight test of an indigenous solid rocket booster in late 2025 mark genuine progress. DRDO's parallel development of a liquid-fuelled ramjet engine for a new long-range supersonic cruise missile programme, which will also benefit BrahMos's future variants, demonstrates the spillover effect of sustained investment in propulsion technology.

On seekers and guidance, the successful testing of an indigenous active radar homing seeker for BrahMos in March 2023 was a milestone. Sustaining this capability and extending it to newer seekers with AESA technology for BrahMos-NG requires continued investment in electronic warfare and signal processing capability within the DRDO-private sector ecosystem.

On hypersonics, BrahMos-II, conceptualised as a Mach 7–8 scramjet-powered missile with significantly extended range, represents the next frontier. DRDO's April 2025 ground test of a scramjet combustor for 1,000 seconds⁵⁸ is encouraging. However, as BrahMos Aerospace CEO Dr Joshi acknowledged, hypersonic development requires time and sustained investment that the current programme structure cannot accelerate beyond a certain pace. The institutional framework for BrahMos-II must be established now.

6.4 | Enabling Defence Exports and Global Positioning

India's emergence as a significant arms exporter, with exports reaching ₹38,424 crore in FY 2025–26, and a target of ₹50,000 crore by 2029⁵⁹, depends critically on the BrahMos programme, which serves as its flagship export. The programme's export success builds institutional capital that can be leveraged for other Indian defence platforms: it demonstrates India's ability to deliver complex systems on time, provide operator training and maintenance support, and honour after-sales commitments.

The government-to-government (G2G) export model developed for the Philippines, featuring diplomatic negotiations, favourable financing, and after-sales support packages built into the deal structure, should be institutionalised as the template for all major Indian platform exports.

This requires: a dedicated defence export promotion organisation within the MoD with sufficient authority and resources; an export credit facility (analogous to the US Ex-Im Bank) to provide competitive financing to buyer nations; and a streamlined regulatory pathway for G2G exports that does not impose the same timelines as commercial procurement.

Section 6

From Programme to Ecosystem: Scaling Strategic Capability

6.5 | Extending the Model to Emerging Domains

The BrahMos framework, comprising a government-backed JV, co-development with a technology partner, a long-term procurement commitment, progressive indigenisation, and MSME ecosystem building, is applicable to multiple emerging capability domains. Three merit particular attention:

01 | Aero-Engine And Propulsion

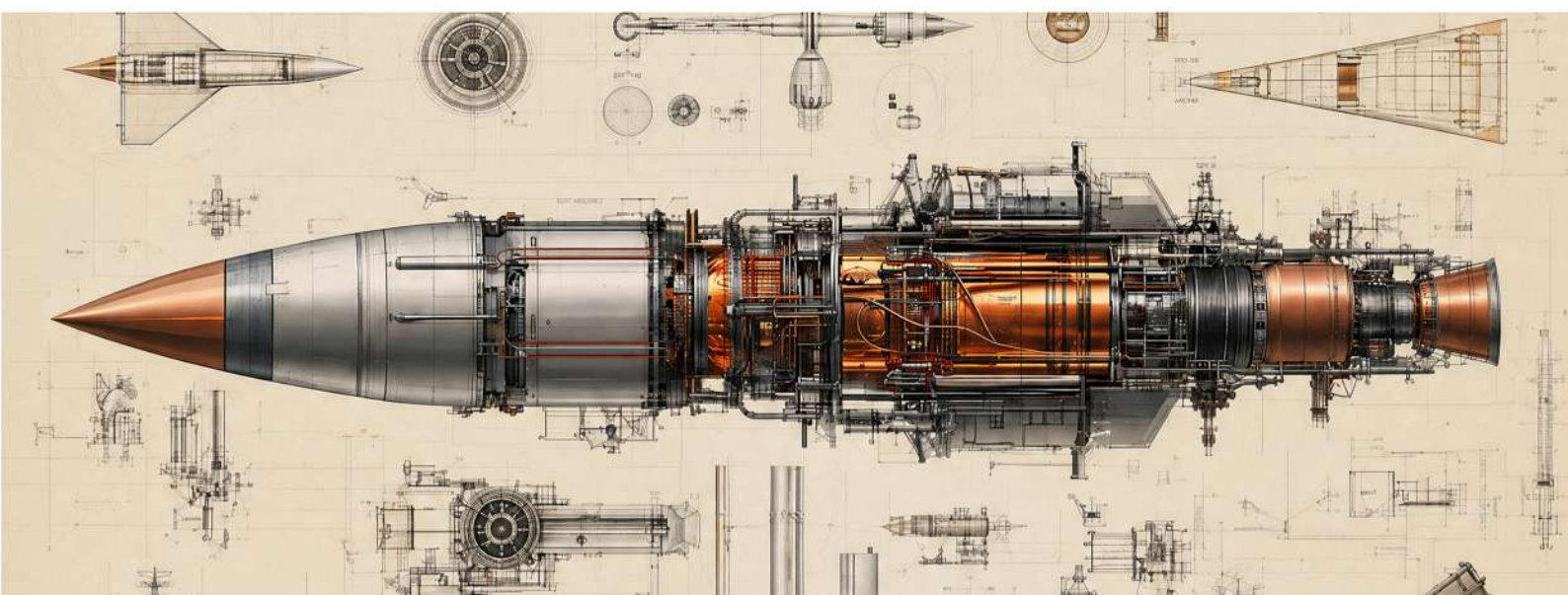
The AMCA engine programme under the GTRE-Safran co-development framework is the most urgent application. As PM Modi stated in his Independence Day 2025 speech, India must develop sovereign engine capability. The institutional structure being assembled, comprising GTRE, IAF, CEMILAC, academic institutions, and the private sector (Tata Group for airframes), mirrors the BrahMos model. The critical missing element is a production-oriented JV entity, analogous to BrahMos Aerospace, that insulates programme execution from DRDO's laboratory governance culture.

02 | Advanced Avionics And Radar

India's reliance on imported radar seekers, electronic warfare systems, and avionics for its major platforms constitutes a capability vulnerability, analogous to the pre-BrahMos dependence on imported cruise missile technology. Establishing a dedicated JV or production entity in this domain, potentially in collaboration with Israel or France, would address this gap.

03 | Unmanned Systems

Loitering munitions, autonomous underwater vehicles, and high-altitude long-endurance UAVs represent the next generation of precision-strike and ISR capability. India's iDEX programme has generated promising private sector innovations. Scaling these to production and integrating them into a dedicated ecosystem development strategy is the next institutional challenge.



Section 7

A National Playbook for Strategic Self-Reliance

The lessons of BrahMos, examined across the preceding sections, converge on a replicable framework for India's next generation of strategic aerospace and defence capability. This section sets out that framework as a six-pillar national playbook, grounded in the programme's experience and applicable to the priority domains identified in Section 6.

The Six-Pillar Playbook at a Glance

Pillar A

**Mission-Oriented
Programme Design
and Governance**

Pillar B

**Co-Development
Over Procurement**

Pillar C

**Long-Term Industrial
Partnerships**

Pillar D

**Technology Absorption
and R&D Continuum**

Pillar E

**Industrial Ecosystem
Development**

Pillar F

**Differentiated
Procurement Approaches
for Strategic Systems**

Section 7

A National Playbook for Strategic Self-Reliance

Pillar A: Mission-Oriented Programme Design and Governance

Strategic capability programmes must be governed as missions, not procurements. A mission-governance framework has four structural elements: first, a dedicated programme entity with a commercial legal form (JV company or equivalent), rather than a government laboratory or a DPSU project; second, empowered leadership with a mandate and tenure measured in years, not annual postings; third, a board accountable for outcomes, with both government and private-sector representation; and fourth, a multi-year budget commitment that cannot be disrupted by annual appropriations cycles.

The BrahMos IGA, signed by both governments, provided the foundational commitment that freed programme management from year-to-year institutional uncertainty. For future major domestic programmes, an equivalent commitment mechanism, with multi-year budget authority, incorporated in the Cabinet Committee on Security sanction note, is required.

National priority domains should be identified explicitly: propulsion and aero-engines; avionics, radar, electronic warfare and mission systems; hypersonic systems; advanced materials for aerospace; next-generation naval systems. Each domain should have an empowered mission structure.

Pillar B: Co-Development over Procurement

The BrahMos model's most important structural feature, and the element most poorly replicated in other Indian defence programmes, is genuine co-development rather than technology transfer or procurement. Co-development means that both parties contribute to the design, share the IP, and build a shared understanding of the system's architecture.

This requires shifting from a 'buy' or 'license-build' procurement model to a 'co-develop and co-own IP' model. Strategic partnerships with global OEMs must include explicit IP provisions granting India the right to modify, improve, and export the co-developed system without restriction. The GTRE-Safran collaboration on the AMCA engine, with full Indian IP ownership, represents the right model. Offset obligations, which have generally failed to deliver meaningful technology transfer, should be restructured as co-development obligations with specific IP deliverables.

Pillar C: Long-Term Industrial Partnerships

National capability cannot be built on transactional procurement relationships. Private-sector enterprises that supply BrahMos, including Godrej Aerospace, HAL's integration division, L&T's defence unit, and PTC Industries, invested in capability over the years in anticipation of sustained programme demand. This investment logic requires relationship continuity that competitive tendering for individual procurement cases cannot provide.

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A National Playbook for Strategic Self-Reliance

A Long-Term Industrial Partnership (LTIP) framework, analogous to the UK's 'Team Complex Weapons' model⁶⁰, in which a prime contractor consortium maintains a long-term relationship with the MoD across multiple programmes, would provide the demand certainty private-sector investment requires. LTIPs should be structured with explicit indigenisation roadmaps, IP co-development obligations, and escalating Indian design and engineering leadership throughout the relationship. Contract renewal should be based on performance against indigenisation milestones, not merely cost and delivery.

Pillar D: Technology Absorption and R&D Continuum

BrahMos's most valuable long-term contribution is not the missile itself but the engineering knowledge embedded in India's defence industrial base, built over 25 years of co-development. This knowledge in precision manufacturing, propulsion chemistry, guidance software, and materials science forms the foundation on which BrahMos-NG, BrahMos-II, and future indigenous programmes will be built.

Institutionalising the R&D continuum from laboratory to production requires three structural elements: DRDO–industry technology transition mechanisms (the Technology Development Fund, expanded and better administered, provides this); university–DRDO–industry research consortia for frontier technologies (similar to the US DARPA–university model); and production-readiness review processes that gate programme funding on demonstrated industrialisation, not merely on laboratory demonstration. The Kaveri programme's failure to transition from laboratory prototype to a production-ready engine is the canonical case of what happens without this continuum.

Pillar E: Industrial Ecosystem Development

Anchor investments in large-scale, sustained procurement programmes build industrial ecosystems by providing demand certainty that cascades through the supply chain. The Lucknow BrahMos facility is the anchor investment building the UP DIC's defence manufacturing cluster. Future anchor investments, one per strategic domain per DIC node, should be identified and committed to within the Defence Industrial Corridor Investment Framework.

MSME integration requires dedicated support mechanisms: technology transfer from DRDO to MSMEs via the Technology Development Fund; quality certification support (AS9100, NADCAP) through government-backed schemes; access to test infrastructure via shared facilities in the Defence Industrial Corridors; and export promotion via the BrahMos export model, under which Indian MSMEs supplying BrahMos components also gain access to global aerospace supply chains. As Godrej & Boyce's Kaustubh Shukla noted: "Exports have been very promising, ranging from 20 per cent to 40 per cent of our defence revenues." This global integration is the ultimate measure of ecosystem maturity.

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A National Playbook for Strategic Self-Reliance

Pillar F: Differentiated Procurement Approaches for Strategic Systems

The core structural reform required is the creation of a differentiated procurement pathway for high-technology strategic programmes, one that explicitly separates the governance logic of these programmes from that of standard commercial procurement.

This policy brief proposes a three-tier taxonomy under DAP 2020 (or its successor, DAP 2026, now under development). Tier 1 covers standard procurement: commercial-off-the-shelf equipment, standard military hardware, and items on the Positive Indigenisation Lists. The standard DAP framework applies, with L1 selection and competitive tendering. Tier 2 covers advanced capability acquisition: platforms and systems requiring significant indigenous design and engineering. The Strategic Partnership Model, enhanced with explicit IP provisions and LTIP mechanisms, applies. Tier 3 covers national capability programmes: the BrahMos-class programmes that are foundational to India's sovereignty. These are governed by dedicated mission structures, co-development JV frameworks, and long-term budget commitments at the Cabinet Committee on Security level. They are not subject to competitive tendering and are governed by negotiated partnership agreements with explicit IP, indigenisation, and industrial development obligations.

This three-tier structure does not require abandoning competitive procurement for most defence acquisitions. It simply recognises that a framework designed for buying a truck is not appropriate for building a hypersonic missile programme, and creates the institutional space for the latter to be governed appropriately.



Section 8

Interviews

Dr V.K. Saraswat

Former DG, DRDO; Former Member, NITI Aayog

Q: Why was the BrahMos programme conceived, and what strategic gap was it meant to fill?

A: BrahMos emerged as a direct response to a specific capability deficit in India's missile ecosystem. The Integrated Guided Missile Development Programme, launched in 1983, created domestic strength in ballistic missiles, surface-to-air missiles, and related systems, but India still lacked a credible cruise missile capability. At that stage, the country had two choices: build such a capability entirely on its own over a much longer timeline, or partner with the one country that then possessed leading supersonic cruise missile expertise, namely Russia. The second path was chosen because India wanted not simply a missile, but a state-of-the-art supersonic cruise missile with immediate strategic relevance. BrahMos was therefore conceived as a strategic capability-building initiative rather than a routine procurement decision.

Q: What made the BrahMos model fundamentally different from a standard buyer-seller or licence-production arrangement?

A: Its uniqueness lies in co-creation. BrahMos was not a conventional transaction in which one side sold technology and the other assembled it. It was a joint venture between two countries with different technological traditions and industrial systems. Responsibilities for subsystems, manufacturing, and production were distributed across both sides. That mattered because it taught India how to collaborate on advanced systems at a strategic level and absorb capability more deeply. BrahMos created production facilities in India, built scale manufacturing capacity, and helped the country internalise expertise in propulsion, guidance, control, and systems integration. Most importantly, it showed that India could manufacture a high-technology defence product at international quality and at a meaningful scale.

Q: How did BrahMos influence India's approach to defence exports and industrial confidence?

A: BrahMos had an impact at three levels: military capability, industrial confidence, and export credibility. It became a tri-service weapon with operational relevance across land, sea, and air roles. At the same time, it forced India to close technology gaps in key areas and demonstrated that the country could build a globally competitive strategic system. Once that credibility was established, interest began emerging from other countries. That changed India's export mindset. Earlier, defence exports were often treated as strategically risky or institutionally undesirable. BrahMos helped push India toward a more structured and confident export posture, while also showing that export ambition can be balanced with safeguards, selective partnerships, and protection of critical technologies.

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Q: What changed in India's indigenous defence ecosystem as a result of broader reforms?

A: A major shift occurred when India stopped relying only on state-owned production structures and began deliberately leveraging the private sector. Private firms, MSMEs, and startups have increasingly moved from component supply into design, development, and manufacturing roles. Mechanisms such as the Development cum Production Partner model, defence industrial corridors in places such as Chennai and Lucknow, offsets, iDEX, and wider access to DRDO test facilities all helped strengthen the ecosystem. These reforms created stronger manufacturing clusters and a more entrepreneurial technology culture. However, the deeper challenge remains industrial R&D. Too many firms are still more comfortable with manufacturing execution than with genuine product creation and long-cycle research.

Q: What should India prioritise over the next 25 years if it wants genuine strategic sovereignty?

A: India must build future capability around co-creation rather than licence dependence. It should expand production scale across key defence categories, because recent conflicts have shown that inventories matter. At the same time, future systems must become more intelligent, autonomous, networked, and cyber-resilient. That means sustained investment in artificial intelligence, robotics, quantum-based navigation, resilient communications, and advanced weapon systems. India also needs stronger risk-sharing and long-tenor financing for industrial R&D, including support through mechanisms such as the Research, Development and Innovation scheme. The central lesson of BrahMos is clear: strategic partnerships should help India capture not only manufacturing know-how, but also the deeper know-why needed to design, improve, and evolve complex systems independently.

Q: How should India structure future collaboration?

A: It must be mission-driven, future-oriented, and anchored in co-development, not assembly. Collaboration should bring together industry, academia, DRDO, national laboratories, and the armed forces around clearly defined critical technologies with shared responsibility, funding, and long-term accountability. That is how India can prepare for the non-contact wars of the future with confidence.

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Interviews

Vikas Khitha

Former VP & Head (Corporate Office & International Relations), L&T Precision Engineering & Systems

Q: What lessons can India draw from strategic programmes such as BrahMos?

A: The most important lesson is that strategic capability is built through sustained collaboration between government institutions, research organisations, the armed forces, and industry. BrahMos showed that when policy continuity, technological ambition, and industrial participation remain aligned over time, India can create a globally competitive capability. Its significance lies not only in the missile itself, but in the industrial ecosystem it created across design, manufacturing, integration, testing, and supply-chain development. That is the broader template India should study as it plans future programmes in aerospace, propulsion, advanced materials, and emerging defence technologies, because enduring capability comes from institutions and ecosystems, not from isolated projects or one-off procurement decisions alone.

Q: Which technology areas should India prioritise to strengthen its aerospace and defence manufacturing base?

A: Recent geopolitical developments and the changing nature of warfare point to four major priorities: integrated air defence systems, combat drones and unmanned platforms, advanced propulsion technologies, and emerging non-kinetic capabilities. Yet the most strategic priority remains aero-engine development. India has made meaningful progress in missiles, aircraft platforms, and space systems, but propulsion remains one of the few critical areas where external dependence persists. Aero engines demand mastery of advanced materials, metallurgy, precision manufacturing, thermodynamics, and systems integration that underpin the broader aerospace ecosystem. If India develops high-thrust indigenous engines for next-generation combat platforms, it will not only reduce dependence but also strengthen the full aerospace ecosystem and move the country closer to genuine technological self-reliance.

Q: What are the principal barriers to greater private-sector participation in defence manufacturing?

A: The key barrier is the lack of long-term visibility and predictability in procurement. Aerospace and defence manufacturing requires substantial investment in infrastructure, technology, certification, quality systems, and highly specialised talent. Companies can justify these investments only when they have confidence in future demand. Many firms are willing to increase their role, but they need clearer signals on production volumes, procurement

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timelines, and programme continuity. Predictability matters even more in high-technology sectors, where capability development can take years and requires major upfront investment. Long-term procurement planning and sustained programme commitments would encourage deeper industrial participation and help build stronger domestic manufacturing capacity.

Q: Does India's procurement framework need to evolve from cost optimisation to capability creation?

A: India's procurement framework has an important role in ensuring transparency, competition, and fiscal discipline. However, as the country moves into complex areas such as aero engines, advanced missile systems, avionics, and next-generation aerospace platforms, procurement policy must increasingly support capability creation alongside cost efficiency. In strategic technologies, technical maturity, indigenous intellectual property, reliability, and industrial capability often carry greater long-term value than the lowest initial price. If procurement gives overwhelming weight to cost once minimum technical criteria are met, it can discourage innovation and weaken investment in frontier technologies. For India's long-term strategic interests, procurement must help build world-class capability, not merely reward the cheapest acceptable bid.

Q: How can India strengthen its defence export ambitions over the next decade?

A: India's defence exports have grown but sustaining that trajectory will require a broader ecosystem approach. Many prospective markets in Southeast Asia, Africa, and Latin America are sensitive to financing support. Leading defence exporters back their industries through sovereign financing, export-credit facilities, and other government-supported financial instruments. India will need a stronger export-financing architecture, but that alone will not be enough. Diplomatic engagement, reliable after-sales support, resilient supply chains, and long-term strategic partnerships are equally important. As India expands into global aerospace and defence markets, it must also navigate sanctions regimes, technology controls, and compliance requirements carefully. Real objective should be to create a globally competitive industrial ecosystem that supports both national security and long-term economic growth.

Q: How important are talent and research institutions in building an aerospace manufacturing base?

A: Human capital is fundamental. Aerospace manufacturing depends on strong capability in engineering, materials science, electronics, propulsion, software, & advanced manufacturing. India, therefore, needs closer collaboration between universities, research institutions, government laboratories, and industry. Creating pathways that connect students & researchers to strategic technology programmes early will strengthen the talent pipeline, accelerate innovation, and support the long-term industrial depth required for globally competitive capability in areas such as propulsion, materials science and AI. It is also, fundamentally, an education and research challenge.

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Interviews

Dr Sudhir Mishra

Former MD & CEO, BrahMos Aerospace

Q: During your tenure at BrahMos Aerospace, what were the most significant milestones achieved by the programme?

A: Several milestones during my tenure materially strengthened India's deterrence capability. One of the most important was the successful integration of BrahMos with the Sukhoi Su-30MKI, creating the world's first supersonic cruise missile system of its kind for an operational fighter aircraft. That achievement greatly expanded the missile's operational flexibility. Another major development came after India joined the Missile Technology Control Regime in 2016, which enabled the range to move beyond the earlier 290-kilometre limit. We also developed a steep-dive capability and advanced different operational variants. On the export side, the agreement with the Philippines in 2022 was a watershed moment because it represented India's first major weapon export contract of this scale and strategic significance.

Q: How did BrahMos evolve from a joint venture into a symbol of Aatmanirbharta?

A: BrahMos evolved through sustained technological advancement and industrial deepening. Initially, the programme combined Russian expertise in certain propulsion-related areas with Indian strengths in systems integration and indigenous subsystems. Over time, however, it grew into a much wider industrial consortium in which industry was treated not merely as a vendor but as a production partner. That distinction was crucial. The programme helped establish a missile industry consortium involving more than 200 public- and private-sector partners. Indigenous work in solid propulsion, advanced guidance, non-metallics, and avionics steadily improved range, accuracy, and reliability. In that sense, BrahMos became more than a missile programme; it became a practical example of how "Make in India" can evolve into "Make for the World."

Q: What were the toughest technological and organisational challenges in scaling the programme?

A: The integration of a heavy missile onto a fighter aircraft without compromising performance or safety was among the toughest technical challenges. We had to design a dedicated airborne launcher and an emergency gas-ejection system, and we supported this work through wind tunnel testing at NAL and detailed computational fluid dynamics analysis. There were also lifecycle challenges, including extending missile life beyond original timelines. Organisationally, the bigger challenge was building and managing an ecosystem that could manufacture, test, store, and support such a complex strategic system over time. Success depended not only on DRDO and BrahMos Aerospace but also on disciplined coordination across multiple industrial partners, laboratories, and users.

Section 8

Interviews

Q: What broader lessons does the BrahMos journey offer for future strategic programmes?

A: The first lesson is that major strategic capability requires political support, stable funding, and mission continuity over many years. The second is that international collaboration can be highly productive if it is used to build domestic expertise rather than prolong dependence. The third is that private industry must be involved early and treated as a real partner in execution and scale-up. Finally, strategic systems should be developed with adaptability in mind. BrahMos proved its value because it could serve land, sea, and air platforms and continue evolving with new variants. That flexibility, combined with reliability, is what converts a successful programme into a long-term national capability.

Q: What do you see as the next frontier for the BrahMos family of systems?

A: The next major step is BrahMos-NG. This lighter and more compact version will allow aircraft to carry more missiles in a single mission while retaining the speed and lethality associated with the BrahMos family. Beyond that lies the hypersonic BrahMos, expected to operate at speeds of Mach 7 and above. Such advances will drastically reduce reaction time for adversaries and open a new chapter in deterrence capability. They will also stimulate further Indian innovation in propulsion, materials science, electronics, and advanced manufacturing. In that sense, the future of BrahMos is not just about one missile family; it is about consolidating India's position as a globally competitive aerospace and defence power.

Q: Why does the industrial consortium matter so much?

A: Because strategic strength does not come from one company alone. BrahMos succeeded because public- and private-sector partners were treated as production partners, creating a dependable ecosystem for manufacturing, testing, maintenance, upgrades, and long-term support at scale. It also demonstrated that export credibility rests on product reliability, execution discipline, and sustained support after induction in global markets today.







Recommendations and **Way Forward**

The following recommendations are addressed to specific institutions and stakeholders. They reflect the analytical conclusions of the preceding chapters. They are designed to be actionable within India's current policy context, including the ongoing draft DAP 2026 review and the Year of Reforms mandate.

Recommendations and Way Forward

For the Government and Ministry of Defence

- Establish a National Strategic Capability Programme (NSCP) framework under the Cabinet Committee on Security, designating 3–5 priority strategic domains (starting with aero-engines, aerospace electronics and mission systems, and hypersonics) and committing multi-year budget authority of 10–15 years for each domain.
- Restructure DAP 2020 (or DAP 2026) to create an explicit Tier 3 procurement pathway for NSCP-designated programmes, with mission-governance JV structures, IP co-development provisions, and long-term industrial partnership (LTIP) contracting as standard provisions.
- Mandate that all future major co-development arrangements include explicit IP provisions ensuring Indian rights to modify, export, and sub-license without reference to the foreign partner, modelled on the AMCA engine GTRE-Safran arrangement.
- Establish a Defence Export Promotion Authority with dedicated resources for government-to-government deal facilitation, competitive export credit financing, and after-sales support commitments, to sustain and scale the BrahMos export template.
- Expand the Defence Industrial Corridors' anchor investment strategy, identifying one strategic capability anchor per DIC node with a 10-year procurement commitment and supporting MSME ecosystem development programme.
- Develop a National Aerospace Supply Chain Strategy focused on critical materials, electronics, semiconductors, sensors, propulsion systems and advanced manufacturing technologies to reduce long-term external dependencies in strategic sectors.

For DRDO and the R&D Ecosystem

- Restructure DRDO's mission to explicitly include industrialisation targets alongside research milestones: each major programme should define production-readiness milestones as primary deliverables, not secondary outcomes.
- Create a Technology Transition Mechanism, a dedicated DRDO function responsible for migrating laboratory technologies into commercial production, with industry partnerships, IP structuring, and production financing.

Recommendations and Way Forward

- Expand the Technology Development Fund substantially (from current levels toward ₹1,000–2,000 crore annually) with simplified grant mechanisms for private sector firms engaged in co-development of indigenisation-critical technologies.
- Establish structured university–DRDO–industry research consortia in frontier technology domains, modelled on the DARPA-university model, to build the long-term scientific base for next-generation strategic programmes.

For Industry: Large Companies and MSMEs

- Large firms (Godrej, L&T, TASL, Mahindra Aerostructures, Bharat Forge, Adani Defence, HAL) should establish dedicated Strategic Programme Units, ring-fenced organisational entities with long-term mandates and leadership accountability for specific strategic domain programmes, insulated from short-term commercial pressures.
- Industry should proactively propose Long-Term Industrial Partnership frameworks to MoD, not merely respond to tenders: the BrahMos model's success was built on the depth of relationships that competitive tendering cannot generate.
- MSMEs should treat quality certification (AS9100, NADCAP) and global supply chain relationships as strategic assets: the BrahMos model shows that participation in defence programmes and access to the global aerospace supply chain are mutually reinforcing.
- Industry bodies (CII, FICCI, SIDM) should develop standardised IP frameworks for defence co-development that can be used as the basis for government negotiation with foreign OEM partners, reducing the current ad-hoc approach to IP structuring in major programmes.
- Increase industry investment in R&D and technology development. India's overall Gross Expenditure on Research and Development (GERD) remains at approximately 0.64 per cent of GDP, well below that of leading aerospace and defence economies. Private-sector participation accounts for only around 36 per cent of national R&D expenditure. Indian industry must move beyond a predominantly manufacturing-led approach and invest more aggressively in indigenous research, product development, advanced materials, propulsion technologies, electronics and next-generation defence systems. Long-term technological sovereignty will require industry to emerge not only as a production partner but also as a co-creator of intellectual property and innovation.

Conclusion

BrahMos was conceived in a year when India was emerging from Pokhran-induced international isolation, Russia was navigating post-Soviet economic collapse, and the world's fastest cruise missile was an unfunded laboratory ambition in Moscow. Twenty-five years later, that ambition has become a combat-proven, 80 per cent indigenised, tri-service deployed strategic weapon system that is transforming India's defence export trajectory and reshaping the strategic balance in the Indo-Pacific.

The journey took 25 years, over 200 Indian enterprises, three generations of programme leadership, and the sustained commitment of two governments across multiple political cycles. It produced not merely a missile but a template for how India can build strategic aerospace and defence capability that the market will not create on its own, that DRDO cannot deliver through laboratory development alone, and that procurement of foreign systems cannot substitute.

The BrahMos model's mission-oriented JV governance, genuine co-development, long-term industrial partnership, MSME ecosystem building, and sustained state demand are replicable. India is now applying versions of it to the AMCA engine (GTRE-Safran co-development with Indian IP), to advanced avionics (strategic partnership frameworks), & to hypersonic propulsion (DRDO scramjet programme). The institutional architecture is being built, if not yet completed.

The structural challenge is real. DAP 2020, despite meaningful reforms, continues to apply procurement logic to capability development challenges. DRDO's institutional culture is optimised for research, not production. Private sector firms are willing but under-resourced for the scale of investment that frontier-technology strategic programmes require.

The Long Game

The Kaveri engine programme's four-decade struggle is a standing reminder of what happens when these structural conditions are not addressed.

India's defence production reached ₹1.54 lakh crore in FY 2024-25. Exports climbed to ₹38,424 crore in FY 2025-26. The Philippines has BrahMos batteries deployed. Indonesia has just signed. Vietnam is in final negotiations. Seventeen nations have expressed interest following Operation Sindoor's combat validation. The momentum is real, and the opportunity is significant.

What this moment requires is not celebration, though the 25th anniversary merits it, but institutional ambition. The ambition to build three more BrahMos-scale programmes in the next decade: one in aero-engines, one in advanced avionics, and one in hypersonics. The ambition is to transform DAP 2020 into a framework that distinguishes strategic capability programmes from standard procurement cases. The ambition is to build a defence export apparatus that can convert the post-Sindoor global interest in Indian weapon systems into signed contracts and delivered capability.

BrahMos has shown that India can build the world's best supersonic cruise missile. The next question is whether India can build the institutional ecosystem to make that achievement the rule rather than the exception.

That is the playbook this policy brief has sought to outline. The anniversary of BrahMos is the right moment to commit to it.



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About the Authors



Anurag Bisen,
Senior Fellow, Vivekananda International Foundation

Captain Anurag Bisen is a veteran Indian Navy submariner with more than 35 years of service. A graduate of Defence Services Staff College Wellington, he holds a Master's degree in Defence and Strategic Studies and Telecommunication & Electronic Warfare.

Captain Bisen is also a Bachelor of Laws (LLB) with a diploma in Russian Language from Arkhangelsk State University, Russia. He has been nominated as a permanent expert on the panel of the International Expert Group of the Lomonosov Moscow State University Marine Research Centre, Russia. He is also Honorary Adjunct Faculty at the National Maritime Foundation, New Delhi.



Siddhartha D. Kashyap,
Adfactors PR

A public affairs professional and former journalist with close to three decades of experience, he advises Indian and global clients on market access, regulatory strategy and stakeholder engagement across sectors including aerospace and defence, energy and infrastructure.



Kritika Khatwani,
Adfactors PR

A policy and communications professional with close to six years of experience, she supports corporations, industry associations and start-ups in navigating policy and regulatory developments, stakeholder engagement and advocacy initiatives across healthcare, pharmaceuticals, agriculture, technology, FMCG and sustainability sectors.



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The Vivekananda International Foundation is an independent non-partisan institution that conducts research and analysis on domestic and international issues and offers a platform for dialogue and conflict resolution. Some of India's leading practitioners from the fields of security, military, diplomacy, government, academia and media have come together to generate ideas and stimulate action on national security issues.

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Contact

Vivekananda International Foundation

3, San Martin Marg, Chanakyapuri, New Delhi

Phone No: +91-(0)11-24121764, +91-(0)11-24106698

Fax No: +91-(0)11-66173415

E-mail: [info\[@\]vifindia.org](mailto:info[@]vifindia.org)

www.vifindia.org

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Contact

Siddhartha D. Kashyap

Phone No: +91 99714 55999

E-mail: siddhartha.kashyap@adfactorspr.com

Kritika Khatwani

Phone No: +91 86048 01332

E-mail: kritika.khatwani@adfactorspr.com